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Effects of K and Mn promoters over Fe_2O_3 on Fischer–Tropsch synthesis

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ABSTRACT

Structural and compositional design of core-shell structure is an effective strategy towards enhanced catalysis. Herein, amorphous MnO_2 nanosheets and K^+-intercalated layered MnO_2 nanosheets are controllably assembled over Fe_2O_3 spindles, in which the MnO_2 nanosheets are perpendicularly anchored to the surface of Fe_2O_3. Such a core shell structure contributes to a high specific surface area and abundant pore channels on the surface of catalysts. In addition, the existence of K^+ provides large numbers of basic sites and restrains the formation of unpleasant $(Fe_{1-x}Mn_x)_3O_4$. Benefiting from the merits in structure and composition, CO adsorption is enhanced and remaining time of intermediates is prolonged on the surfaces of catalysts during the Fischer–Tropsch synthesis (FTS), facilitating to the formation of active iron carbides and C–C coupling reactions. Resultantly, the Fe_2O_3@ K^+-MnO_2 shows both a high CO conversion of 82.3% and a high C_{5+} selectivity of 73.1%. The present study provides structural and compositional rationales on design high-performance catalysts towards FTS.

1. Introduction

With rapid depletion of limited fuels and chemicals used in transportation industry, Fischer–Tropsch synthesis (FTS) on converting syngas (a mixture of CO and H_2) to liquid hydrocarbon fuels has attracted increasing attentions [1,2]. Fe, Co and Ru with proper phases show appreciable activity for FTS process [3]. Among them, Fe-based catalysts are promising candidates due to the merits of low cost, high availability and high resistance to contaminants [3–7]. In addition, Fe-based catalysts present a high activity for the water-gas shift reaction, which enables the catalyst adaptive to syngas with a wide range of H_2/CO ratios [5–7]. However, improvement in the CO conversion and selectivity for desired hydrocarbons is in urgent need for Fe-based catalysts [3,4]. Especially, enhancing selectivity for C_{5+} hydrocarbons with a high CO conversion is of great importance because the main components in gasoline are C_5–C_{12} hydrocarbons [8,9].

During the FTS process, Fe_2O_3 is generally reduced to a mixture of low-valence iron oxides and iron carbides by syngas [6,7]. The iron carbides are the active phases, in which CO is adsorbed, dissociated and finally converted to hydrocarbons by surface polymerization reactions [2,4,10–12]. Therefore, selectivity and activity of Fe_2O_3 are highly related to its surface structure, and composition [13].

Manganese and potassium are widely used as the promoters to improve the FTS performance by tailoring the surface structure and composition of Fe-based catalysts [7,9,14–16]. However, the effects of manganese and potassium promoters on the performance of Fe_2O_3 catalysts are complicated: (1) Effects of Mn and K promoters are different, which mean that surface chemistry is very important in determining the activity of catalysts. Manganese promotes dispersion of iron and accelerates reduction of Fe_2O_3 to FeO [14,15]. However, formation of active iron carbide phases is simultaneously suppressed after addition of manganese due to the formation of unpleasant mixed oxides $(Fe_{1-x}Mn_x)_3O_4$ [14]. Instead, formation of iron carbides can be promoted by addition of potassium due to enhanced CO chemisorption, contributing to increased selectivity for C_{5+} hydrocarbons [16]. (2) Configurations between promoters and Fe_2O_3 also affect the performance of catalysts. A highly porous surface structure is beneficial for adsorption of gaseous reactants (CO and H_2) and prolonging the remaining time of intermediate products. Also, MnO_2 existing as coated layers on the Fe_2O_3 spindle surfaces shows a much higher CO conversion than that as $Fe_{3-x}Mn_xO_4$ [8,10,14]. (3) The content of introduced promoters is another factor determining the performance of cata-

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lysts. A proper amount of promoter could significantly improve the activity of iron-based catalysts while excess amounts of promoters tend to block the active sites on the catalyst surface, causing the deactivation of catalysts [16].

Elaborately tuning the configuration between promoters and Fe_2O_3 to maximize the surface exposure of active sites is an effective way to enhance the FTS performance. In addition, by co-incorporation of manganese and potassium promoters, the positive roles of both two promoters are likely to be exploited by a synergistic effect, contributing to fabricating high-performance iron-based catalysts.

In this paper, amorphous MnO_2 nanosheets and K^+-intercalated layered MnO_2 nanosheets are controllably assembled over Fe_2O_3 spindles matrix, in which the MnO_2 nanosheets are perpendicularly anchored to the surface of Fe_2O_3. The obtained Fe_2O_3@amorphous MnO_2 and Fe_2O_3@ K^+-MnO_2 are then used as the model architectures to probe the effects of K and Mn promoters on FTS. With this special structure, abundant micro channels are formed in the shell, facilitating the adsorption and circulating of gaseous molecules. It is found that the Mn promoter significantly increases CO conversion but slightly decreases C_{5+} selectivity. Further incorporation with the K promoter enhances both CO conversion and C_{5+} selectivity. The Fe_2O_3@ K^+-MnO_2 therefore shows a high CO conversion of 82.3% and a high C_{5+} selectivity of 73.1%. The present study provides structural and compositional rationales on design high-performance catalysts towards FTS.

2. Experimental

2.1. Preparation of Fe_2O_3@MnO_2 spindles

Firstly, 2.703 g $FeCl_3 \cdot 6H_2O$ (analytical grade, Aldrich) and 0.0351 g $NaH_2PO_4 \cdot H_2O$ (analytical grade, Aldrich) were dissolved in 500 mL deionized water to form a clear solution and then the mixture was sealed in a blue-cap bottle. After that, the bottle was heated to 105 °C in a blast air oven and kept for 48 h. Finally, the Fe_2O_3 powders were obtained after centrifugation, washed with deionized water and ethanol for several times, and finally vacuum-dried at 70 °C overnight.

0.2 g $KMnO_4$ was dissolved in 25 mL deionized water. After sonication for 10 min, 0.4 mL HCl (0.2 M) was added into the solution. Then, 90 mg Fe_2O_3 spindles were dispersed in 25 mL deionized water by sonication for 10 min. After that, the $KMnO_4$ solution and the Fe_2O_3 dispersion was mixed and reacted for 10 h at 95 °C in an oil bath. The Fe_2O_3@MnO_2 (Fe_2O_3@MnO_2-95) was then obtained after centrifugation, washed with deionized water and ethanol for several times, and finally vacuum-dried at 70 °C overnight. Similarly, Fe_2O_3@MnO_2-140 was prepared in a similar way by changing the reaction temperature to 140 °C and duration to 2 h. The hybrids obtained at 95 and 140 °C are denoted as Fe_2O_3@MnO_2-95 and Fe_2O_3@MnO_2-140, respectively. For comparison, the sample of Fe_2O_3-K is prepared by mixing the obtained Fe_2O_3 spindles with 4.0 wt% K_2CO_3 (equal to 2.3 wt% K).

2.2. Characterization

The crystalline structures of the as-obtained powders were characterized by X-ray diffraction (XRD) using a PANalytical X'Pert Pro X-ray diffractometer with Ni filtered Cu $K\alpha$ radiation (λ=1.5406 Å). X-ray photoelectron spectroscopy (XPS) was performed on a VG ESCALAB210 apparatus with Mg $K\alpha$ source. The C 1*s* as a reference signal was adjusted to 284.8 eV. The CasaXPS software was used to obtain the high-resolution spectra envelopes by using symmetric Gaussian-Lorentzian product functions to approximate the line shapes of the fitting curves. The structures and morphologies of samples were investigated by using a field emission scanning electron microscopy (FESEM) and a JEM-2100 transmission electron microscopy (TEM) equipped with an Energy Dispersive Spectrometer (EDS). The surface areas and pore features of the fresh catalysts were measured at −196 °C using a N_2 physisorption instrument (Micromeritics ASAP2020). The surface area and pore size distribution were determined by Brunauer–Emmett–Teller (BET) and Barrett–Joyner–Halenda (BJH) methods. The compositions of the samples were also analyzed by Inductively Coupled Plasma Optical Emission Spectrometer (ICP-OES, Optima 5300DV). Thermogravimetric analysis (TGA) of the spent catalysts was carried out in air atmosphere using a Mettler Toledo DSC1 TGA apparatus at a heating rate of 10 °C min^{-1} from room temperature to 850 °C.

H_2-Temperature-Programmed-Reduction (H_2-TPR) was carried out on Micrometrics Auto Chem II 2920 using a mixture of 5% H_2/95% Ar (v/v) gas with a flow of 50 mL min^{-1}. Before the experiment, 0.1 g catalyst was pretreated in He stream (40 mL min^{-1}) at 120 °C for 2 h. A thermal conductivity detector (TCD) was used to monitor the variation of gas concentration.

H_2 or CO temperature-programmed desorption (H_2-TPD or CO-TPD) was also carried out at the same chemisorption analyzer. Firstly, pretreatments of the catalysts were carried out by loading 0.1 g catalysts into the U-type quartz tube reactor in a nitrogen (N_2) gas atmosphere at a flow rate of 50 mL min^{-1} until the spectra became stable. To convert the original catalysts to active iron carbide species, the catalysts were then activated at 300 °C for 10 h in the 50% H_2/50% CO (v/v) gas at a flow of 3000 mL g^{-1} h^{-1}. After this process, iron carbide is massively produced (as revealed by the XRD patterns of activated Fe_2O_3@MnO_2-140 in Fig. S1), proving the successful activation of original catalysts. Finally, TPD was carried out after H_2 or CO adsorption on the samples lasted for 1 h at 40 °C with a flow of 50 mL min^{-1}. Considering that only one CO molecule can be adsorbed by one active iron site [17–19], the active sites per gram catalysts ($Fe_{active\ site}$) can be determined from the amount of chemisorbed CO ($CO_{chemisorbed}$, µmol g^{-1}) in the CO-TPD curves. CO_2 temperature-programmed desorption (CO_2-TPD) is also conducted by the same chemisorption analyzer to quantitatively determine the basic sites of catalysts.

2.3. Catalytic activity towards FTS

The Fischer–Tropsch synthesis reaction was conducted in a fixed-bed reactor. Typically, 0.5 g catalyst and 0.5 g quartz sand were well mixed to avoid agglomeration. Prior to the FTS reaction, the samples were blasted with N_2 to refresh the surfaces. When the temperature was increased to 120 °C, the gas was switched to H_2 at a flow of 3000 mL g^{-1} h^{-1} and the temperature was subsequently heated to 350 °C with a ramp of 1 °C min^{-1}. Then, the catalysts were in-situ reduced at 350 °C for 10 h under a H_2 flow of 3000 mL g^{-1} h^{-1}. After the above pre-activation process, the reaction was performed for 30 h at 280 °C and 2.0 MPa in the syngas of 50% H_2/50% CO (v/v, 3000 mL g^{-1} h^{-1}). The tail gas was analyzed by a gas chromatograph (FULT 979011) equipped with a TCD and FID (flame ionization detector). After the reaction, the collected liquid products were analyzed off-line by the gas chromatograph (FULT 979011). Correspondingly, the product selectivity and reactivity of the catalysts were calculated.

The iron-time-yield (FTY, mol_{co} g_{Fe}^{-1} s^{-1}), which means the conversion amount of CO in one second per gram iron, are calculated by the following equation:

$$\mathrm{FTY} = \frac{n_{\mathrm{CO}}}{m_{\mathrm{Fe}} \times t} \tag{1}$$

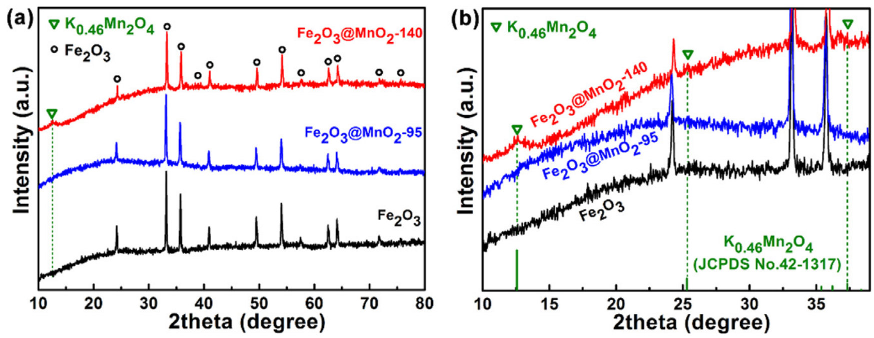

Fig. 1. (a) XRD patterns of the Fe_2O_3, $Fe_2O_3@MnO_2$-95 and $Fe_2O_3@MnO_2$-140; (b) a magnified view of (a).

where n_{co} (mol) is the conversion amount of CO, m_{Fe} (g) is the mass of iron element and t (s) is the reaction time.

The catalytic reaction rate of Fe-based catalysts can be evaluated by the turn over frequency (TOF, s^{-1}) according to the following equation [17]:

$$\mathrm{TOF} = \frac{\mathrm{CO}_{\mathrm{reaction\ rate}}}{\mathrm{Fe}_{\mathrm{active\ site}}} \tag{2}$$

where $\mathrm{CO}_{\mathrm{reaction\ rate}}$ (μmol g^{-1} s^{-1}) is the average conversion rate of CO per gram catalysts while $\mathrm{Fe}_{\mathrm{active\ site}}$ (μmol g^{-1}) is the active sites per gram catalysts, which can be calculated by reactions (3) and (4), respectively:

$$\mathrm{CO}_{\mathrm{reaction\ rate}} = \frac{\mathrm{CO}_{\mathrm{in}} \times \eta_{\mathrm{CO}}}{22.4 \times 3600} \times 1000 \tag{3}$$

$$\mathrm{Fe}_{\mathrm{active\ site}} = \mathrm{CO}_{\mathrm{chemisorbed}} \tag{4}$$

where $\mathrm{CO}_{\mathrm{in}}$ (mL g^{-1} h^{-1}) is the CO flow rate in the syngas (1500 mL g^{-1} h^{-1}); η_{CO} is the CO conversion ratio; Assuming that only one CO molecule can be adsorbed by one active iron site [17], then $\mathrm{Fe}_{\mathrm{active\ site}}$ is equal to the amount of chemisorbed CO ($\mathrm{CO}_{\mathrm{chemisorbed}}$, μmol g^{-1}), which can be derived from the CO-TPD results.

3. Results and discussion

3.1. Morphology and composition characterizations

The crystallographic configurations of the as-prepared samples are identified. As shown in Fig. 1(a), the XRD pattern of the prepared Fe_2O_3 matches well with hematite phase (JCPDS No. 33-0664, hexagonal, R-3C, $a = b = 5.04$ Å, $c = 13.75$ Å, $\alpha=\beta=90°$, $\gamma=120°$). Similarly, the hematite is also the dominant phase in $Fe_2O_3@MnO_2$-95 and $Fe_2O_3@MnO_2$-140, without detecting MnO_2 phase, indicating that MnO_2 in the sample is amorphous [20]. For the $Fe_2O_3@MnO_2$-140, a new peak at 12.5° appears, which in conjunction with other two weak peaks (Fig. 1(b)) are well indexed to $K_{0.46}Mn_2O_4$ (JCPDS No. 42-1317, monoclinic, C2/m, $a = 5.15$ Å, $b = 2.844$ Å, $c = 7.159$ Å, $\alpha=90°$, $\beta= 100.64°$, $\gamma=90°$). It should be mentioned that $K_{0.46}Mn_2O_4$ is only observed in $Fe_2O_3@MnO_2$-140, indicating that chemical-bonded K^+ is formed by increasing the reaction temperature. Higher temperatures, e.g. 140 °C, facilitate formation of crystalline layered MnO_2, in which K^+ is intercalated between interlayer gaps of layered MnO_2 to form $K_{0.46}Mn_2O_4$.

Perpendicularly anchored MnO_2 layers on the surface of Fe_2O_3 spindles are verified by comparing the morphologies of Fe_2O_3, $Fe_2O_3@MnO_2$-95 and $Fe_2O_3@MnO_2$-140 (Fig. 2). Uniform and well dispersed pristine Fe_2O_3 with spindle-like morphology are prepared by adding PO_4^{3-} (NaH_2PO_4) as surfactants (Fig. 2(a) and (d)) [21]. Herein, the molar ratio between Fe^{3+} and PO_4^{3-} is 44.6. The observed lattice fringe is 0.27 nm, corresponding to (104)

Table 1. Compositions of the samples determined by ICP.

Sample	Fe (wt%)	Mn (wt%)	K (wt%)	Fe/Mn (wt/wt)	K/Mn (wt/wt)
$Fe_2O_3@MnO_2$-95	56.6	11.8	0.1	4.79	0.03
$Fe_2O_3@MnO_2$-140	54.2	12.5	2.3	4.40	0.19

plane of hematite Fe_2O_3 (Inset in Fig. 2(d)) [21]. In acid solution, MnO_4^- can be reduced to MnO_2 particles [22]. After introducing MnO_4^-, MnO_2 shell is successfully coated while the spindle-like morphology remains (Fig. 2(b) and (e)), indicating Fe_2O_3 spindles can act as an effective scaffold for MnO_2 growth. Specially, the MnO_2 shell consists of nanosheets [23], which are anchored to the surface of Fe_2O_3 and perpendicular to the main axis of the Fe_2O_3 spindles. No isolated MnO_2 nanosheets appear. Such a highly ordered architecture forms a porous structure and maximizes the exposure of both Fe_2O_3 and MnO_2 surfaces. As shown in Fig. 2(c) and (f), the $Fe_2O_3@MnO_2$-140 presents a similar structure as that of the $Fe_2O_3@MnO_2$-95. The MnO_2 nanosheets in the shell of $Fe_2O_3@MnO_2$-140 are thinner than that in the $Fe_2O_3@MnO_2$-95 (Fig. 2(e) and (f), insets in Fig. 2(b) and (c)). As illustrated in Fig. 2(g), thinner MnO_2 nanosheets provides abundant pore channels between MnO_2 sheets, facilitating the adsorption of gaseous reactants (CO and H_2) and prolonging the remaining time of intermediate products. In addition, the interlayer space between MnO_2 nanosheets in the $Fe_2O_3@MnO_2$-140 is larger than that in the $Fe_2O_3@MnO_2$-95, facilitating the fully exposure of inner Fe_2O_3 species.

The surface structure and chemistry are further revealed by elements distribution and EDS analysis. Well-defined boundaries between Fe and Mn distributions occur in both $Fe_2O_3@MnO_2$-140 (Fig. 3(a)) and $Fe_2O_3@MnO_2$-95 (Fig. S2), which verifies that Mn exists in the shell and Fe exists in the core. In addition, K element distributed in the shell is only observed in $Fe_2O_3@MnO_2$-140 (Fig. 3(a) and Fig. S2), further indicating that formation of $K_{0.46}Mn_2O_4$ (chemical bonded K^+) only takes place at the higher temperature of 140 °C. Compared with $Fe_2O_3@MnO_2$-95, much enhanced intensity in K spectrum in Fig. 3(b) is also observed for $Fe_2O_3@MnO_2$-140, further suggesting that K^+ can be only chemical-bonded into MnO_2 by increasing the temperature.

The above results are further testified by the ICP results in Table 1. The K content in the $Fe_2O_3@MnO_2$-140 is 2.3 wt%, which is 23 times higher than that (0.1 wt%) in the $Fe_2O_3@MnO_2$-95. The trace amount of K in the $Fe_2O_3@MnO_2$-95 is likely attributed to the physical-adsorbed K^+ while the much higher K content in the $Fe_2O_3@MnO_2$-140 is due to the chemical-bonded K^+ in $K_{0.46}Mn_2O_4$ (Fig. 1). The mass ratios of Fe/Mn in the two samples are comparable, making the two hybrids a good couple for highlighting the influence of K element on the catalytic performance. The results

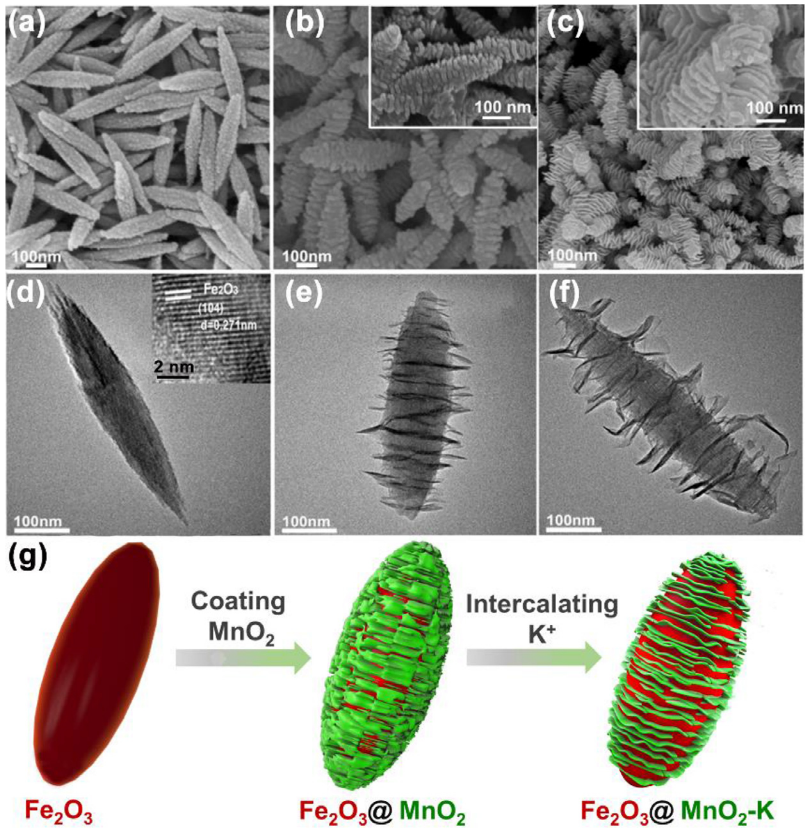

Fig. 2. SEM (a–c) and TEM (d–f) images of the Fe_2O_3 (a, d), Fe_2O_3@MnO_2-95 (b, e) and Fe_2O_3@MnO_2-140 (c, f). (g) The morphology illustration of Fe_2O_3, Fe_2O_3@MnO_2-95 (denoted as Fe_2O_3@MnO_2) and Fe_2O_3@MnO_2-140 (denoted as Fe_2O_3@MnO_2-K).

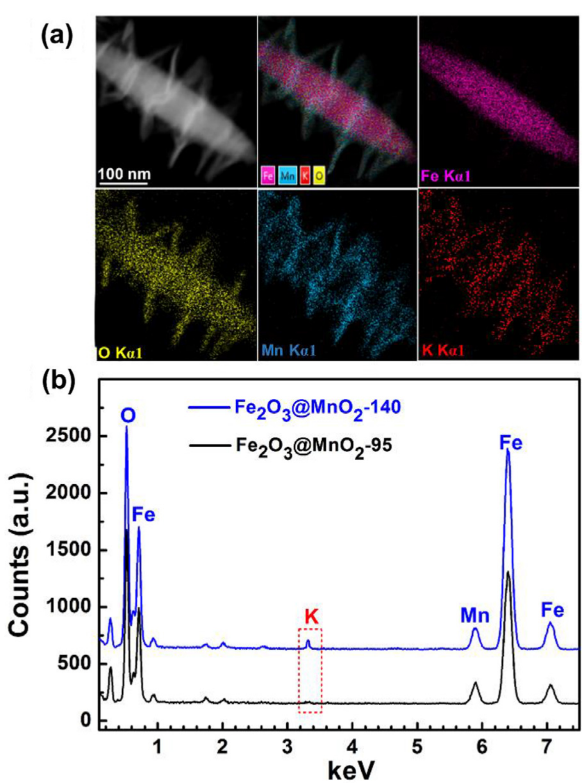

Fig. 3. (a) Element mapping of the Fe_2O_3@MnO_2-140; (b) EDS spectra of the Fe_2O_3@MnO_2-95 and Fe_2O_3@MnO_2-140.

indicate that potassium and manganese promoters are in-situ introduced by increasing the temperature to 140 °C. Therefore, the property difference between Fe_2O_3@MnO_2-95 and Fe_2O_3@MnO_2-140 are highly related to the huge dissimilarity in K content.

The chemistry differences in Fe_2O_3, Fe_2O_3@MnO_2-95 and Fe_2O_3@MnO_2-140 are further uncovered by XPS results. Fig. S3 shows the full XPS spectra of the Fe_2O_3 (black), Fe_2O_3@MnO_2-95 (blue) and Fe_2O_3@MnO_2-140 (red). Signals of Fe 2*p*, O 1*s* and C 1*s* are detected in all samples. For the Fe_2O_3@MnO_2-95 and Fe_2O_3@MnO_2-140, spectra of Mn 2*p* are also observed and the spectra intensity of Fe 2*p* decreases substantially after introducing MnO_2 shell, further verifying that manganese promoters are successfully loaded on the Fe_2O_3 surface. Signals of K 2*p* and K 2*s* are only detected in the Fe_2O_3@MnO_2-140, proving that K can only be loaded by increasing the temperature of hydrothermal reaction. The results are consistent with that in Figs. 1, 3 and Table 1. Fe, Mn and K are also detected in the high-resolution XPS spectra of Fe_2O_3@MnO_2-140, further implying the co-existence of Mn and K on the surface of Fe_2O_3 spindles (Fig. S4). The comparison of O 1*s* spectra in Fig. S5 shows that the surface hydroxyl species (O_H) can be removed by the coated MnO_2 layer (Fe_2O_3@MnO_2-95 in Table S1), which can be further eliminated by intercalation of K^+ (Fe_2O_3@MnO_2-140 in Table S1), indicating that introduced Mn and K can tune the physicochemical characteristics of catalyst surfaces.

The above results show that amorphous MnO_2 nanosheets are perpendicularly anchored onto the surface of Fe_2O_3 spindles. By increasing the reaction temperature from 95 °C to 140 °C, the MnO_2 nanosheets become thinner, providing abundant pore channels (Fig. 2(g)) for adsorption of gaseous reactants and facilitating the fully exposure of inner iron species. In addition, formation of chemical-bonded K^+ is also realized by formation of $K_{0.46}Mn_2O_4$ at 140 °C, resulting in co-incorporation of Mn and K on

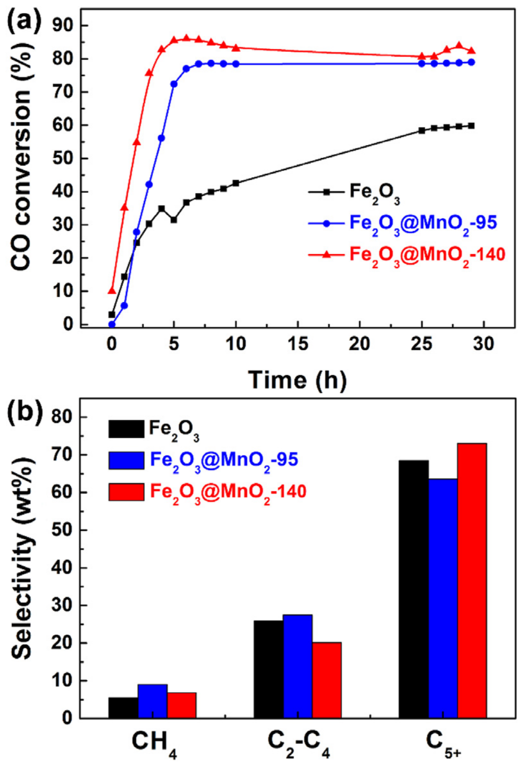

Fig. 4. (a) CO conversion of the Fe_2O_3, Fe_2O_3@MnO_2-95 and Fe_2O_3@MnO_2-140; (b) product distribution.

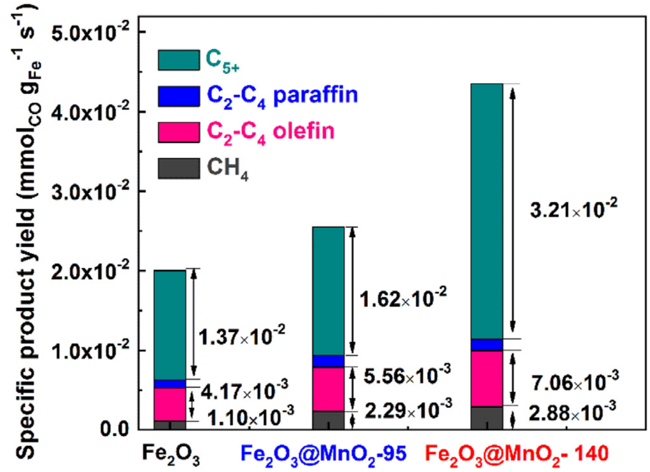

Fig. 5. Specific product yield of the prepared catalysts after 30 h on the syngas stream.

the surface of Fe_2O_3. Such a special porous surface structure and co-incorporation of Mn and K may endow Fe_2O_3@MnO_2-140 outstanding performance for FTS reaction.

3.2. FTS performance analysis

In the initial stage of FTS, Fe_2O_3 is gradually reduced to iron carbides by the syngas. Then, the mixture of H_2 and CO is converted to hydrocarbons on the surfaces of iron carbides [24]. For all samples, CO conversion increases in the initial 6 h and then reaches to an equilibrium value (Fig. 4(a)), implying more and more Fe_2O_3 spindles are reduced to iron carbides. The variation of CO conversion is negligible in the reaction period of 25–29 h for Fe_2O_3 (Fig. S6), indicating that the reaction time (29 h) is enough for evaluating the performance of the catalysts. Compared with Fe_2O_3, the induction periods for Fe_2O_3@MnO_2-95 and Fe_2O_3@MnO_2-140 are significantly shortened, which is likely attributed to the promoted reduction of Fe_2O_3 by Mn and K promoters (Fig. S7 and Table S2). The CO conversions of Fe_2O_3@MnO_2-95 (78.2%) and Fe_2O_3@MnO_2-140 (82.3%) after 30 h are much higher than that of Fe_2O_3 (58.1%). The iron-time-yield (FTY, $mol_{co}\ g_{Fe}{}^{-1}\ s^{-1}$, the conversion amount of CO in one second per gram iron) of the Fe_2O_3 (black), Fe_2O_3@MnO_2-95 (blue) and Fe_2O_3@MnO_2-140 (red) at 30 h reaches to 3.67×10^{-5}, 5.1×10^{-5} and 8.39×10^{-5} $mol_{co}\ g_{Fe}{}^{-1}\ s^{-1}$, respectively (Fig. S8) [25]. Based on the chemisorbed amounts of CO in the CO-TPD results (Fig. S9), the TOF (s^{-1}) values of the Fe_2O_3, Fe_2O_3@MnO_2-95 and Fe_2O_3@MnO_2-140 are calculated to be 6.58×10^{-2}, 7.02×10^{-2} and $9.43 \times 10^{-2}\ s^{-1}$, respectively (Table S3). The outstanding performance of Fe_2O_3@MnO_2-140 in CO conversion may be attributed to the thinner MnO_2 nanosheets and additional K^+ on the surface of Fe_2O_3, which can facilitate the exposure of more active iron species (illustrated in Fig. 2(g)) and tune the surface physicochemical properties of catalysts, respectively. It was reported that when the potassium loading exceeds 0.7 wt%, the activity of catalysts declines sharply due to the blocking of active sites by excess potassium elements [16]. However, the Fe_2O_3@MnO_2-140 possesses a high potassium content of 2.3 wt% and still shows a superior activity. This is likely attributed to the special shell structure of the Fe_2O_3@MnO_2-140, in which the perpendicularly coated MnO_2 nanosheets endures a higher loading amount of potassium without blocking the active sites on the Fe_2O_3 surfaces.

In addition to a high value of CO conversion, Fe_2O_3@MnO_2-140 also possesses the best performance in promoting the selectivity and yield for C_{5+} hydrocarbons. The C_{5+} selectivity of the Fe_2O_3, Fe_2O_3@MnO_2-95, and Fe_2O_3@MnO_2-140 is 68.6 wt%, 63.5 wt%, and 73.1 wt%, respectively (Fig. 4(b)). Correspondingly, the selectivity of C_2–C_4 hydrocarbons is 25.9 wt%, 27.5 wt%, and 20.1 wt%, respectively (Fig. 4(b)). Compared with Fe_2O_3, the decrease in C_{5+} selectivity of Fe_2O_3@MnO_2-95 should be attributed to the formation of unpleasant mixed oxides $(Fe_{1-x}Mn_x)_3O_4$, which can cut off the neighboring active iron sites and block the C–C coupling to long chain hydrocarbons [8,26–31], causing a decreased C_{5+} selectivity and increased C_2–C_4 selectivity. Fe_2O_3@MnO_2-140 shows the highest value of C_{5+} selectivity and lowest value of C_2–C_4 hydrocarbons, which is likely attributed to the K^+-induced increase in CO/H_2 concentration ratio on the catalyst surface [16], benefiting to continued C–C chain growth for formation of heavy hydrocarbons. The specific yields to C_{5+} hydrocarbons of the Fe_2O_3, Fe_2O_3@MnO_2-95 and Fe_2O_3@MnO_2-140 are 1.37×10^{-2}, 1.62×10^{-2} and 3.21×10^{-2} $mmol_{co}\ g_{Fe}{}^{-1}\ s^{-1}$ respectively (Fig. 5), which means that Fe_2O_3@MnO_2-140 also presents the highest yield for production of C_{5+} hydrocarbons. Such results are consistent with the higher values of chain growth probability (α) for Fe_2O_3@MnO_2-95 and Fe_2O_3@MnO_2-140 when compared with that of Fe_2O_3 (Fig. S10).

The FTS experiment results after reaction of 30 h are shown in Table S4. The overall olefins/paraffin mass ratio is increased from 7.49% (Fe_2O_3) to 7.86% (Fe_2O_3@MnO_2-95), and further increased to 8.92% (Fe_2O_3@MnO_2-140). When manganese promoters are loaded, the chain growth is terminated by β-hydride abstraction to form olefins [8]. Potassium promoters can decrease the H_2 concentration on the catalyst surfaces, restraining the hydrogenation process [8]. Consequently, the loading of manganese and potassium promoters leads to a higher olefin/paraffin mass ratio. The distributions of reaction products are shown in Fig. S11. For the Fe_2O_3 (Fig. S11a), the main reaction products are C_2–C_4 olefins. However, the mass ratio of C_{5+} olefin increases remarkably for Fe_2O_3@MnO_2-140 (Fig. S11c), which is also higher than that of Fe_2O_3@MnO_2-95 (Fig. S11b). The mass ratio of oxides in the reaction products of Fe_2O_3 is much

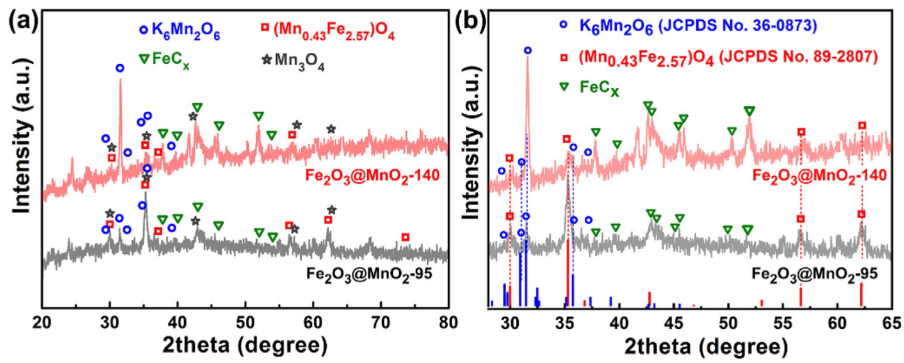

Fig. 6. (a) XRD patterns of the Fe_2O_3@MnO_2-95 and Fe_2O_3@MnO_2-140 after FTS for 30 h on syngas stream; (b) a magnified view of Fig. 6(a).

higher than that of Fe_2O_3@MnO_2-95 and Fe_2O_3@MnO_2-140, indicating that presence of manganese and potassium promoters can restrain the formation of oxides. In comparison, the CO conversion of Fe_2O_3-K is 65.83% (Fig. S12a), which is much lower than that of Fe_2O_3@MnO_2-140 (82.3%, Fig. 4(a)). In addition, the C_{5+} yield of Fe_2O_3@MnO_2-140 is much higher than that of Fe_2O_3-K. The results further manifest the synergistic effect of Mn and K promoters for enhancing the FTS performance. It should be mentioned that the CO conversion at 60 h is almost the same with that at 20 h for Fe_2O_3-K (Fig. S12a), further implying that the reaction period of 29 h is enough for evaluating the performances of catalysts herein.

Conclusively, Fe_2O_3@MnO_2-140 presents the highest values in both CO conversion and C_{5+} selectivity as well as the production yield of C_{5+} hydrocarbons among the samples of Fe_2O_3, Fe_2O_3@MnO_2-95, and Fe_2O_3@MnO_2-140. Such a superior performance of Fe_2O_3@MnO_2-140 is likely attributed to the special surface structure and composition: (1) the ultrathin MnO_2 nanosheets perpendicularly anchored on the surface of Fe_2O_3 can provide abundant pore channels, enhancing the CO adsorption and increasing the remaining time of reaction intermediates. (2) The chemical-bonded K^+ can tune the surface physicochemical properties, which can contribute to an increased CO/H_2 concentration ratio on the catalyst surfaces [15,32,33], benefiting to C–C chain growth for heavy hydrocarbon formation.

3.3. Structure-performance correlation

3.3.1. Critical role of K^+ on enhancing C_{5+} selectivity

The critical role of K^+ on enhancing the FTS performance of Fe_2O_3@MnO_2-140 is embodied by the following aspects: (1) With the help of K^+ ions, the formation of unpleasant $(Fe_{1-x}Mn_x)_3O_4$ is restrained by preferential generation of $K_aMn_bO_c$, contributing to formation of more active iron carbides. (2) Existence of K^+ increases the surface basicity of catalysts, which facilitates the electron transfer from the active iron species to CO, resulting in enhanced C–C coupling reaction [34]. In addition, adsorption of CO in the high-basicity surface can be also strengthened, contributing to an increased CO/H_2 ratio on the surface of catalysts [34,35]. Resultantly, both the enhanced C–C coupling and increased CO/H_2 ratio contribute to formation of heavy hydrocarbons, leading to a high selectivity for C_{5+} products.

Although Mn promoter is commonly used to enhance the performance of iron-based catalysts for Fischer–Tropsch synthesis, Mn can also stabilize the Fe^{2+}/Fe^{3+} ions by formation of mixed oxides $(Fe_{1-a}Mn_a)_3O_4$, which can suppress the further reduction and carburization of catalysts [36–38], hindering the generation of active iron carbide species. The existence of K^+ in Fe_2O_3@MnO_2-140 can restrain the formation of unpleasant $(Fe_{1-x}Mn_x)_3O_4$ by preferential generation of $K_aMn_bO_c$, as evidenced by the fact that $K_{0.46}Mn_2O_4$ is detected in Fe_2O_3@MnO_2-140 (Fig. 1). By keeping the mixture of Fe_2O_3, MnO_2, and K_2CO_3 at 280 °C (the temperature used for FTS reaction) for 30 h in Ar atmosphere, $K_{2-x}Mn_8O_{16}$ (JCPDS No. 44-1386) rather than $(Fe_{1-x}Mn_x)_3O_4$ is detected (Fig. S13), manifesting the preferential formation of $K_aMn_bO_c$ when K^+ is introduced. $K_6Mn_2O_6$ (JCPDS No. 36-0873, monoclinic, P21/b, a = 8.89 Å, b = 6.76 Å, c = 11.39 Å, α=90°, β= 90°, γ=132.13°) is also observed in the activated Fe_2O_3@MnO_2-140 (Fig. S1), further verifying the preferential formation of $K_6Mn_2O_6$ in the conditions for Fischer–Tropsch synthesis. Zhang et al. also demonstrates the easiness for formation of $K_{1.04}Mn_8O_{16}$ between K_2O and MnO_2 [39]. The strong Mn–K interaction in Fe–Mn–K catalyst is also reported by Tao et al. [36]. The intensity of diffraction peaks of $K_6Mn_2O_6$ in the XRD patterns of spent Fe_2O_3@MnO_2-140 is much stronger than that in the spent Fe_2O_3@MnO_2-95 (Fig. 6(a)). Correspondingly, compared with the spent Fe_2O_3@MnO_2-95, the peak intensities of unpleasant $(Mn_{0.43}Fe_{2.57})O_4$ in the spent Fe_2O_3@MnO_2-140 is significantly weakened, accompanied by increased peak intensities of iron carbides. The results suggest that existence of K^+ can restrain the formation of unpleasant $(Fe_{1-x}Mn_x)_3O_4$ by preferential generation of $K_aMn_bO_c$, contributing to production of more active iron carbides. No other K-and Mn-related species are detected in the spent catalysts, revealing that $K_6Mn_2O_6$ is the most stable oxide among all $K_aMn_bO_c$ in conditions for the FTS reactions. The low reaction temperature is likely the reason for the phenomenon that no MnO is detected in the spent catalysts (Figs. 6(a) and S7).

The TEM images of the spent catalysts after FTS for 30 h on syngas stream also reveal the more abundant $K_6Mn_2O_6$ and iron carbides in Fe_2O_3@MnO_2-140 when compared with Fe_2O_3@MnO_2-95 (Fig. 7). $K_6Mn_2O_6$ (with a lattice fringe of 0.28 nm) and Fe_5C_2 (with a lattice fringe of 0.21 nm) are more abundant in the spent Fe_2O_3@MnO_2-140 (Fig. 7(d)) [4]. Instead, Mn_3O_4 (with a lattice fringe of 0.48 nm) and Fe_3O_4 (with a lattice fringe of 0.25 nm) are observed (Fig. 7(c)) [6,23], which present a similar crystal structure with that of $(Fe_{1-x}Mn_x)_3O_4$. The results further indicate that the chemically bonded K^+ in the Fe_2O_3@MnO_2-140 can restrain the formation of unpleasant $(Fe_{1-x}Mn_x)_3O_4$ by preferential generation of $K_6Mn_2O_6$, which can result in more active iron carbides, being consistent with the results in Fig. 6. Based on the hypothesis that only one CO molecule can be adsorbed per active site [17], the active site density for the Fe_2O_3, Fe_2O_3@MnO_2-95 and Fe_2O_3@MnO_2-140 is calculated by reaction (4) to be 0.22, 0.32 and 0.47 $mmol_{CO}/g_{Fe}$ respectively (Fig. S9), further manifesting the increased iron carbide active sites after introduction of K^+ [16]. The overall spindle morphology of the spent catalysts remains intact (Fig. 7(a) and (b)), surrounded by the liquid hydrocarbon products (insets in Fig. 7(a) and (b)). More hydrocarbons on the surfaces of the spent Fe_2O_3@MnO_2-140 suggests a higher selectivity for C_{5+} products, which is further manifested by the highest weight loss for Fe_2O_3@MnO_2-140 in the TGA curves of spent catalysts (Fig. S14).

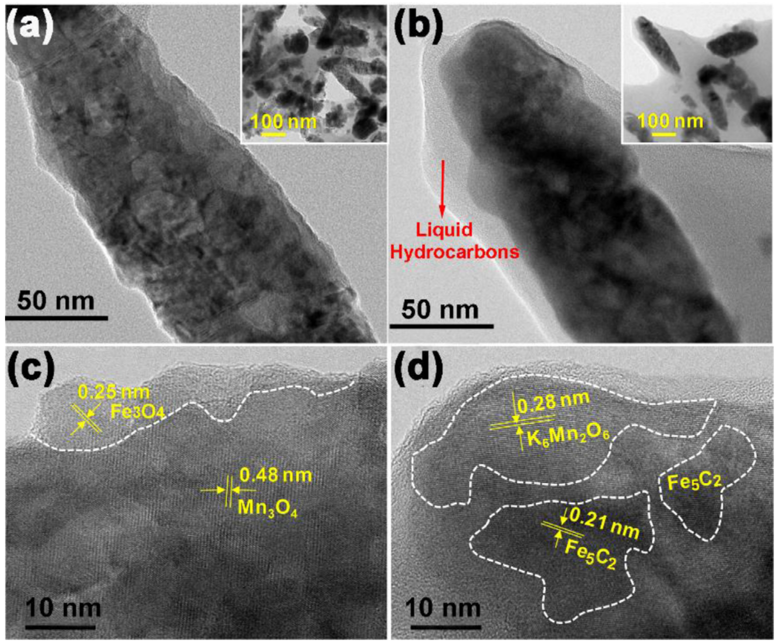

Fig. 7. TEM images of the spent Fe_2O_3@MnO_2-95 (a, c) and Fe_2O_3@MnO_2-140 (b, d) after FTS for 30 h on syngas stream.

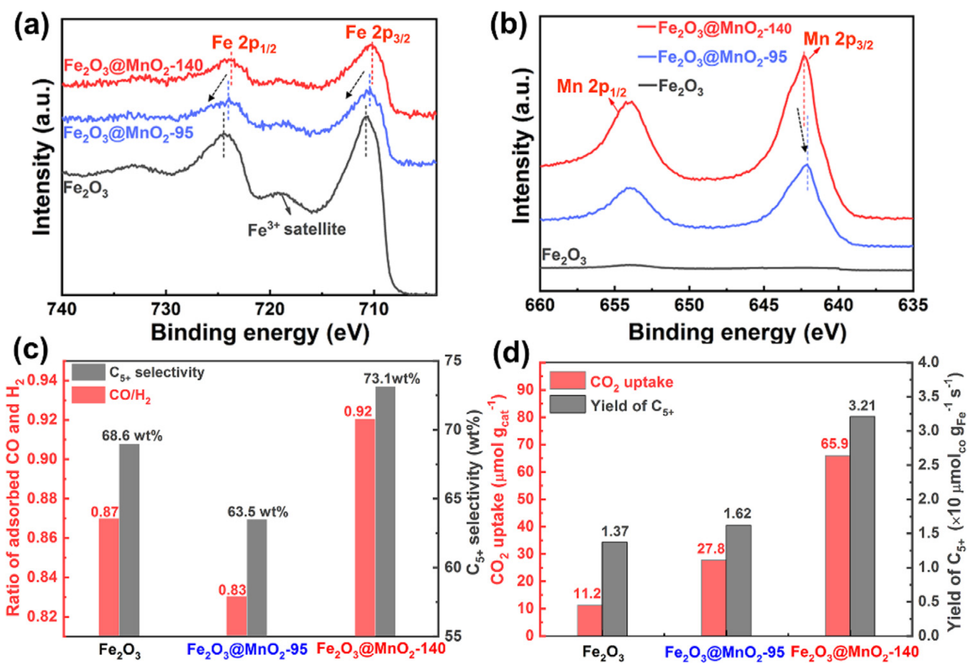

Fig. 8. Comparison of Fe_2O_3, Fe_2O_3@MnO_2-95 and Fe_2O_3@MnO_2-140: high resolution XPS spectrum of (a) Fe 2*p* and (b) Mn 2*p*; (c) Ratio of adsorbed CO and H_2 as well as C_{5+} selectivity; (d) C_{5+} yield and the CO_2 uptake determined by CO_2-TPD method in Fig. S15.

Existence of K^+ can provide abundant basic sites on the surface of catalysts, which can promote the electron transfer from iron species to CO, resulting in enhanced C–C coupling reaction [34,41,42]. The CO_2 uptake determined by CO_2-TPD method (Fig. S15) can reflect the amount of basic site, which is significantly increased by introducing K^+, as evidenced by the CO_2-TPD curves of Fe_2O_3, Fe_2O_3@MnO_2-95, and Fe_2O_3@MnO_2-140 (Fig. S15) [34]. The high-resolution XPS spectrum of Fe 2*p* and Mn 2*p* in Fig. 8(a) demonstrates the electron transfer tendency between Fe_2O_3 and MnO_2. Compared with Fe_2O_3, the Fe 2*p* peak in Fe_2O_3@MnO_2-95 shifts to a lower binding energy (Fig. 8(a)), showing a decrease of approximately 0.3 eV in binding energy. After further introducing K^+ into the MnO_2 shell (Fe_2O_3@MnO_2-140), the decrease of binding energy in Fe 2*p* peak is further enlarged to 0.5 eV. Correspondingly, the Mn 2*p* peak in Fe_2O_3@MnO_2-140 shifts toward a higher binding energy when compared with that in Fe_2O_3@MnO_2-95 (Fig. 8(b)). Such results imply that electrons are transferred from MnO_2 to Fe_2O_3, which is further enhanced by introducing K^+. The enrichment of electrons on the surfaces of iron species can facilitate the adsorption of CO molecules and activate the C–O bonds, enhancing the C–C coupling reactions for promoting formation of heavy hydrocarbon products [38,40]. The ratio of adsorbed CO and H_2 is decreased from 0.87 in Fe_2O_3 to 0.83 in Fe_2O_3@MnO_2-95 (Fig. 8(c)), indicating that MnO_2 layer leads to a lower CO/H_2

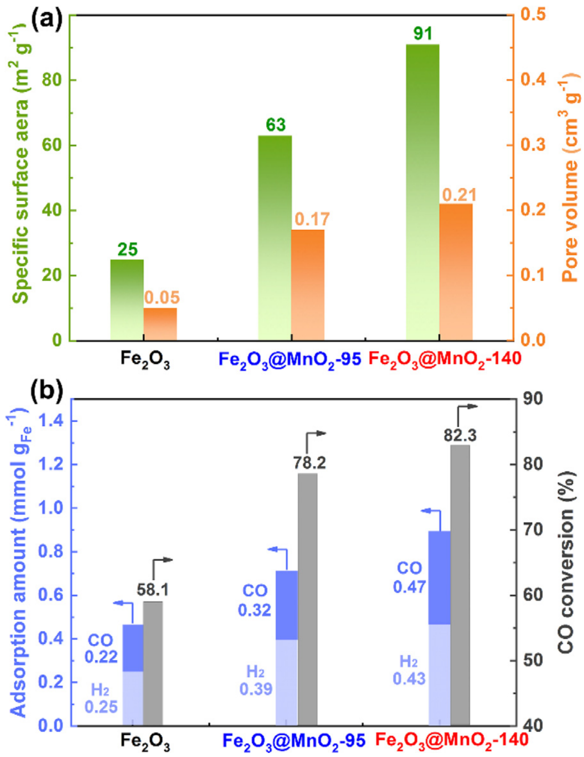

Fig. 9. Comparison of Fe_2O_3, Fe_2O_3@MnO_2-95, and Fe_2O_3@MnO_2-140: (a) specific surface areas and pore volumes; (b) adsorption amount of CO and H_2 as well as CO conversion.

ratio on the surface of catalysts [8,26–31,36–38], which is disadvantageous for generation of heavy hydrocarbons. Interestingly, the value is increased from 0.83 in Fe_2O_3@MnO_2-95 to 0.92 in Fe_2O_3@MnO_2-140 (Fig. 8(c)), implying that the increased basic sites provided by K^+ can result in a high CO/H_2 ratio on the surface of catalyst [34,41]. Specially, the ratio of adsorbed CO and H_2 presents a similar variation trend with that of C_{5+} selectivity (Fig. 8(c)), suggesting that the value of CO/H_2 on the surface of catalysts is likely one of the important reasons determining the selectivity for C_{5+} products. Compared with Fe_2O_3@MnO_2-95, the basic sites (or CO_2 uptake) of Fe_2O_3@MnO_2-140 is significantly increased (Fig. 8(d)), which is in accordance with the greatly enhanced C_{5+} yield of Fe_2O_3@MnO_2-140, further manifesting the important role of basic sites for enhancing the formation of heavy hydrocarbons [42,43].

Therefore, K^+ in Fe_2O_3@MnO_2-140 can restrain the generation of unpleasant $(Fe_{1-x}Mn_x)_3O_4$ by preferential formation of $K_6Mn_2O_6$, contributing to production of more active iron carbides. In addition, the basic sites provided by K^+ in Fe_2O_3@MnO_2-140 facilitate the enrichment of electrons on the surface of iron species, promoting the C–C coupling reactions. Moreover, the abundant basic sites provided by K^+ can result in an increased ratio of adsorbed CO and H_2 on the surface of catalysts, beneficial for formation of heavy hydrocarbons. The above merits result in a high selectivity and yield for C_{5+} products.

3.3.2. Surface construction on the influence of FTS performance

Both the K and Mn promoters and the iron species are essential for enhancing the performance of catalysts. A porous structure can promote the adsorption of gaseous reactants (CO and H_2) while the fully exposure of inner iron species can facilitate the formation of active iron carbides. Compared with Fe_2O_3@MnO_2-95, the thinner MnO_2 nanosheets perpendicularly anchored on the surface of Fe_2O_3 spindles can not only provide a more porous structure (see SEM images in the insets of Fig. 2(b) and (c) and the illustration in Fig. 2(g)), but also maximize the exposure the inner iron species (see Fig. 2(b)–(g)), therefore contributing to better performance for FTS reactions.

The most porous structure on the surface of Fe_2O_3@MnO_2-140 is revealed by the results obtained from Nitrogen adsorption-desorption curves (Fig. S16). The TEM images of Fe_2O_3, Fe_2O_3@MnO_2-95, and Fe_2O_3@MnO_2-140 (Fig. 2(d)–(f)) show a similar size. However, a huge difference in the specific surface area between these three samples is observed (Fig. 9(a)). The specific surface areas of Fe_2O_3@MnO_2-95 and Fe_2O_3@MnO_2-140 are 63 and 91 m^2 g^{-1} respectively, which greatly exceed that of Fe_2O_3 (25 m^2 g^{-1}). The results imply that Fe_2O_3@MnO_2-95 and Fe_2O_3@MnO_2-140 possess a highly porous surface structure, which is also revealed by the morphologies in SEM (insets in Fig. 2(b) and (c)) and TEM images (Fig. 2(e) and (f)). Specially, the specific surface area of Fe_2O_3@MnO_2-140 (91 m^2 g^{-1}) is 1.44 times higher than that of Fe_2O_3@MnO_2-95 (63 m^2 g^{-1}), indicating more abundant pores exist on the surface of Fe_2O_3@MnO_2-140, which is evidenced by the comparison of pore volumes (Fig. 9(a)).

Compared with Fe_2O_3@MnO_2-95, the more abundant pores on the surface of Fe_2O_3@MnO_2-140 contribute to a higher amount of

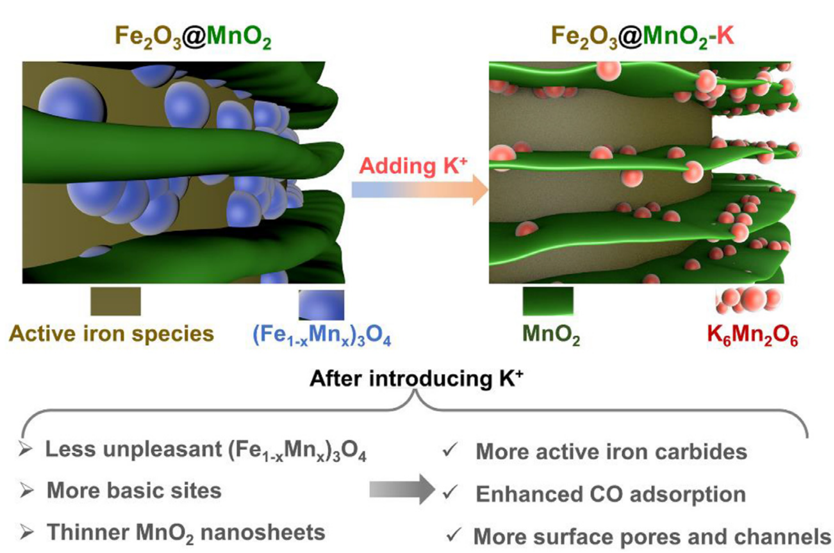

Fig. 10. Illustration of surface structures for activated Fe_2O_3@MnO_2-95 and Fe_2O_3@MnO_2-140.

adsorbed gaseous reactants (CO and H_2, as shown in Fig. 9(b)), resulting in a higher CO conversion (Fig. 9(b)). It should be mentioned that the MnO_2 contents on the surfaces of $Fe_2O_3@MnO_2$-140 and $Fe_2O_3@MnO_2$-95 are almost the same (Table 1), however, the specific surface area in $Fe_2O_3@MnO_2$-140 is 1.44 times that of $Fe_2O_3@MnO_2$-95, accompanied by significantly increased pore volume in the former case (Fig. 9(a)). The results imply that more surfaces of inner Fe_2O_3 core are exposed in $Fe_2O_3@MnO_2$-140, making the contact of reactants (CO and H_2) with the active iron species more easily. Moreover, the abundant pore channels on the surface of $Fe_2O_3@MnO_2$-140 can prolong the remaining time of intermediate light hydrocarbons, facilitating the C–C coupling reactions for formation of heavy hydrocarbons. Therefore, both the CO conversion (Fig. 9(b)) and C_{5+} selectivity (Fig. 8(c)) of $Fe_2O_3@MnO_2$-140 are higher than that of Fe_2O_3 and $Fe_2O_3@MnO_2$-95.

The special surface physicochemical properties of $Fe_2O_3@MnO_2$-140, which means a highly porous structure and co-incorporation of Mn and K promoters, results in a high CO conversion and superior selectivity for C_{5+} hydrocarbons (illustrated in Fig. 10). Specifically, the existence of K^+ restrains the generation of unpleasant $(Fe_{1-x}Mn_x)_3O_4$ by preferential formation of $K_6Mn_2O_6$, resulting in production of more active iron carbides. In addition, the abundant basic sites provided by K^+ in $Fe_2O_3@MnO_2$-140 contributes to a high CO/H_2 ratio on the surface of catalysts, facilitating the C–C chain growth reactions. Moreover, the highly porous structure and abundant pore channels can not only enhance the adsorption of gaseous reactants, but also prolong the remaining time of reaction intermediate, beneficial for both CO conversion and C–C coupling reactions.

4. Conclusions

$Fe_2O_3@MnO_2$ spindles with a porous core-shell structure were prepared, in which MnO_2 nanosheets were perpendicularly anchored to Fe_2O_3 surfaces. The special porous MnO_2 shell structure on the Fe_2O_3 surface promotes the adsorption of syngas and facilitates the reduction of Fe_2O_3, contributing to an increased CO conversion. By increasing preparation temperature, intercalation of chemical-bonded potassium promoter into the MnO_2 shell was realized. Further intercalation of K can decrease the thickness of MnO_2 nanosheets and increase the interlayer space, maximizing the exposure of active sites. The chemically bonded K^+ in the MnO_2 shell can promote the formation of active iron carbides and increase the CO/H_2 concentration ratio on the catalyst surfaces, enhancing the chain growth of carbon for C_{5+} formation. Resultantly, benefiting from the synergistic effect of Mn and K promoters, both the C_{5+} selectivity and CO conversion are increased for the $Fe_2O_3@MnO_2$-K.The present study highlights the effectiveness of core-shell structures with elaborately designed configurations for advanced catalysis.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Supporting information

Elements mapping, nitrogen adsorption/desorption isotherms, pore-size distributions, XPS results, iron-time-yield, XRD patterns, TPD curves, TGA curves, reaction products distribution, specific surface area, pore volume, molar ratios of different oxygen atoms on surfaces and temperatures of reduction peaks for different samples are available.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.jechem.2019.12.003.

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